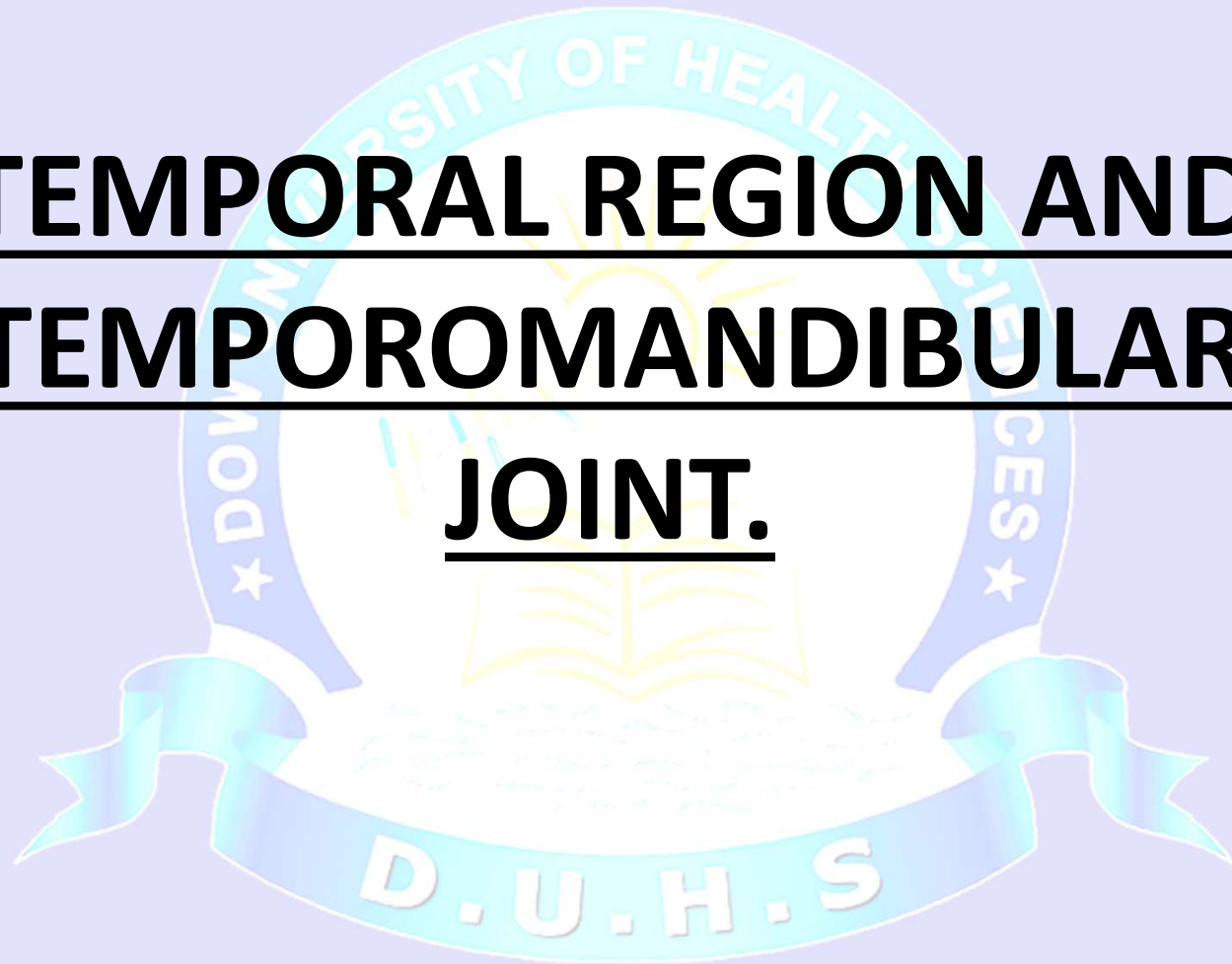


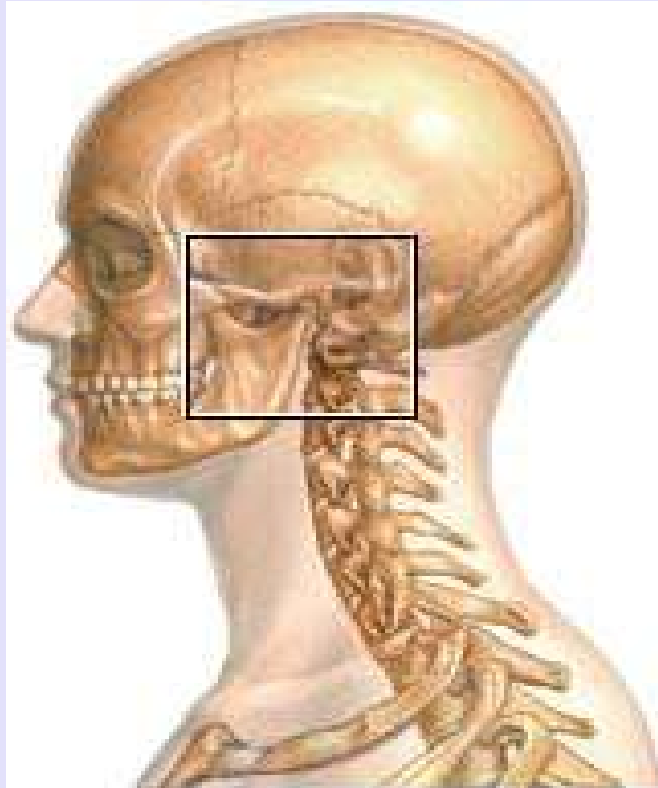
TEMPORAL REGION AND **TEMPOROMANDIBULAR** **JOINT.**



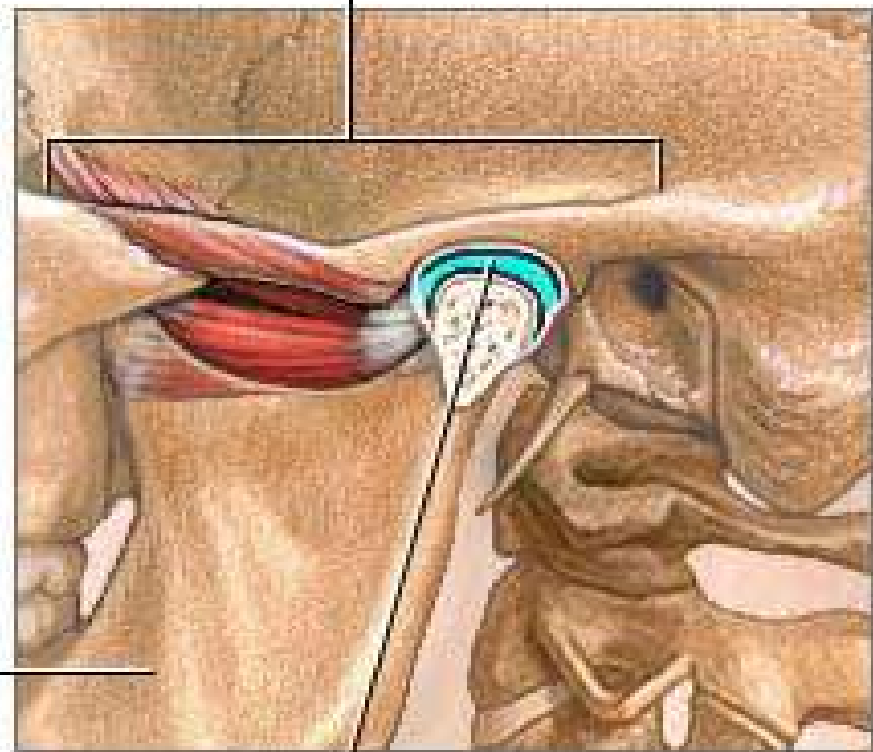
LEARNING OBJECTIVES.

BY THE END OF THE LECTURE, THE STUDENTS SHOULD BE ABLE TO KNOW:

- THE BOUNDARIES AND CONTENTS OF TEMPORAL FOSSA.
- THE TEMPOROMANDIBULAR JOINT, ITS TYPE, FORMATION, NEUROVASCULAR SUPPLY AND ACTION.
- CLINICAL CORRELATES RELATED TO THE TEMPOROMANDIBULAR JOINT.



Zygomatic arch (cheekbone)



Mandible

Temporomandibular joint

Temporal region

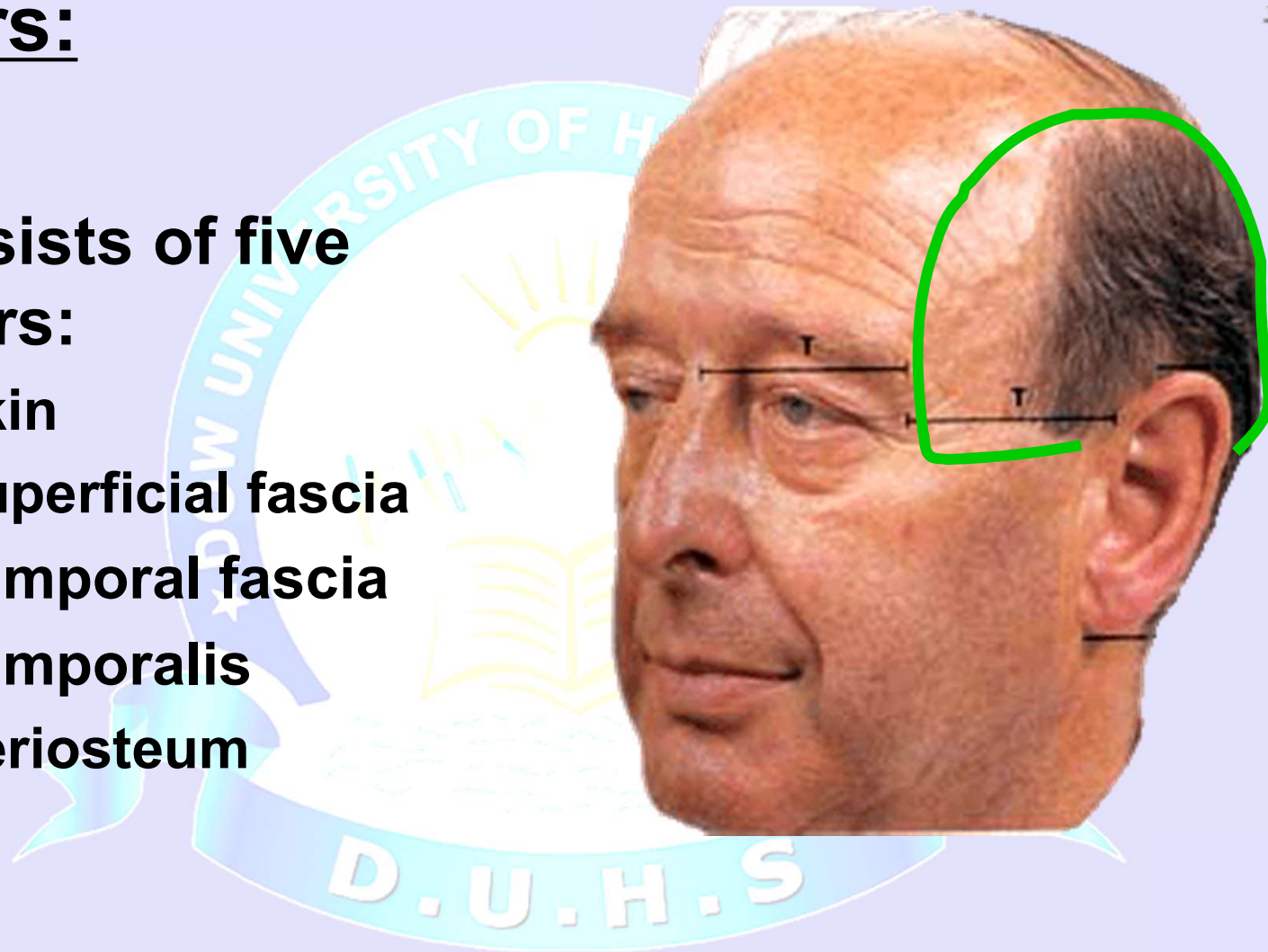
Boundaries :

- Superior: superior temporal line.
- Inferior: superior border of zygomatic arch
- Anterior: the junction of frontal and zygomatic bones
- Posterior: posterior portion of superior temporal line



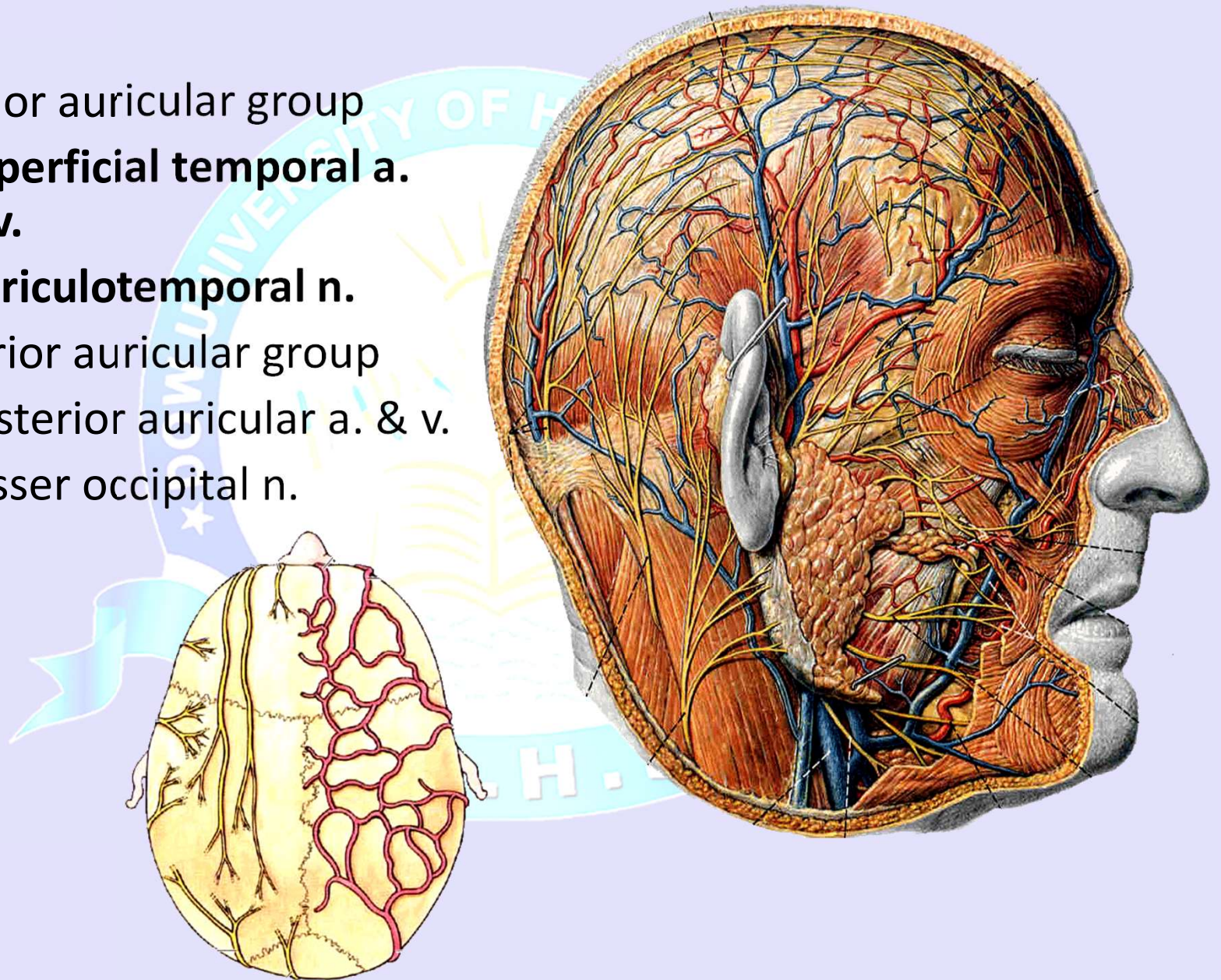
Layers:

- consists of five layers:
 - Skin
 - Superficial fascia
 - Temporal fascia
 - Temporalis
 - Periosteum

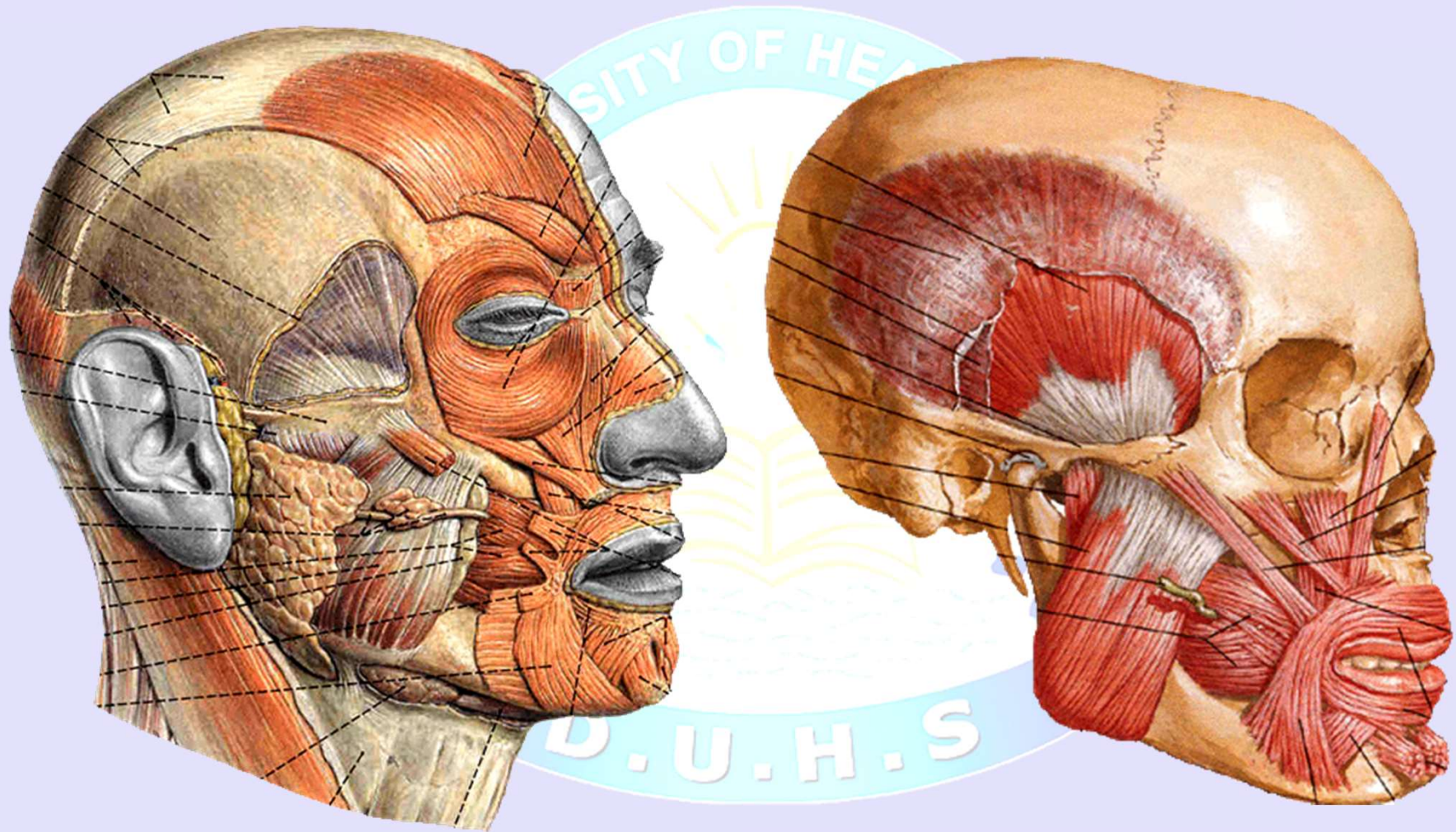


Superficial fascia:

- Anterior auricular group
 - **Superficial temporal a. & v.**
 - **Auriculotemporal n.**
- Posterior auricular group
 - Posterior auricular a. & v.
 - Lesser occipital n.

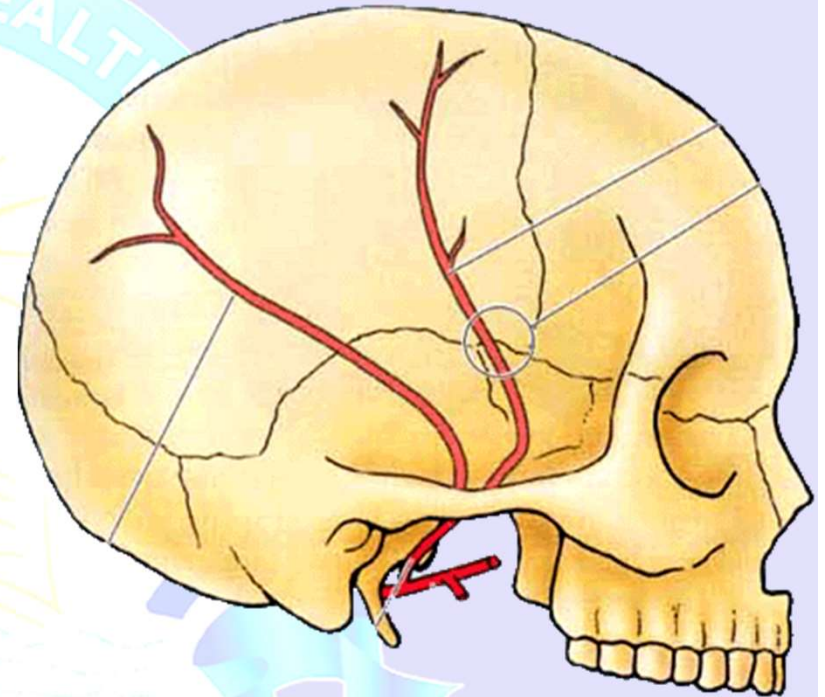


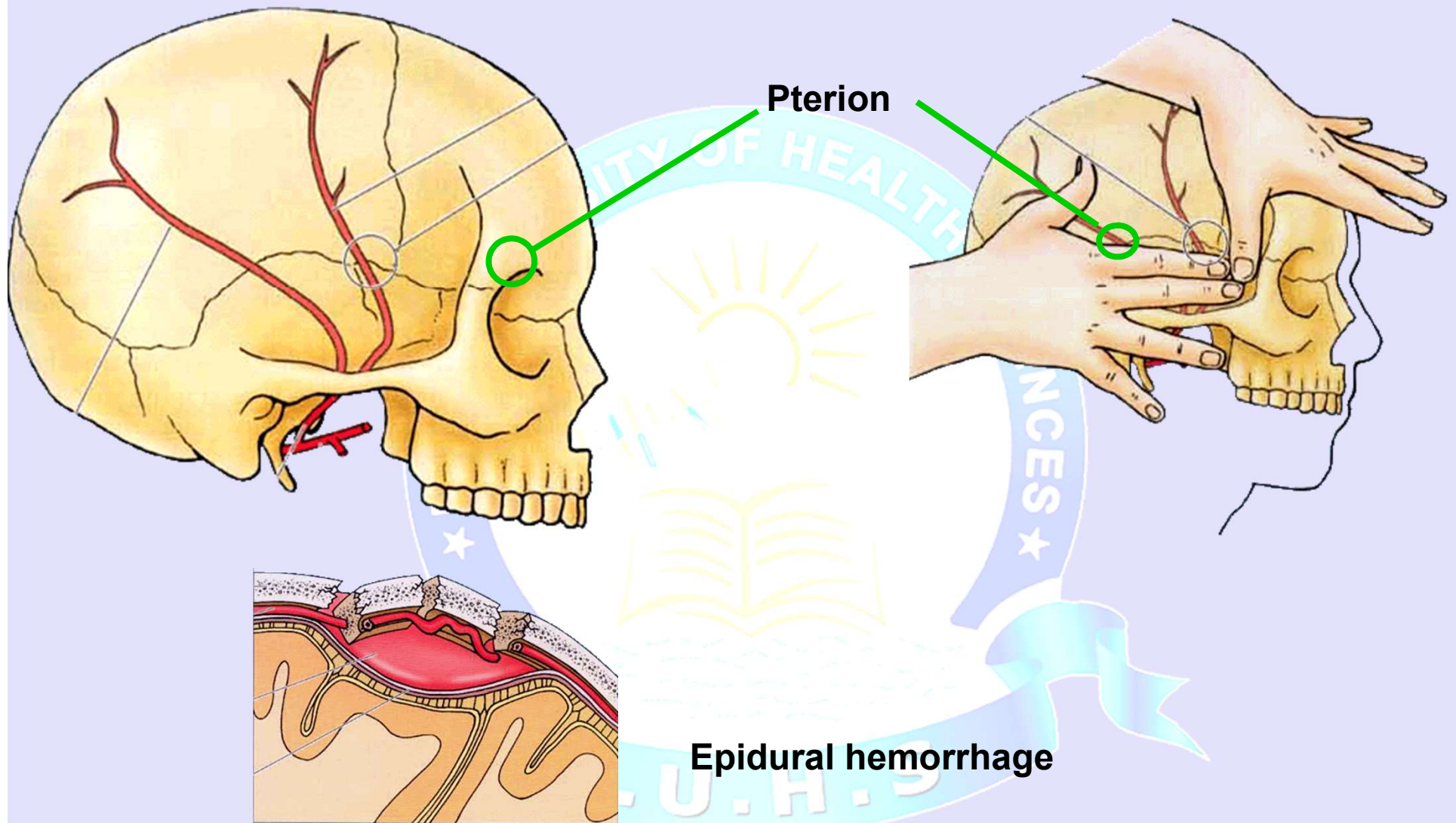
Temporal fascia



Pterion.

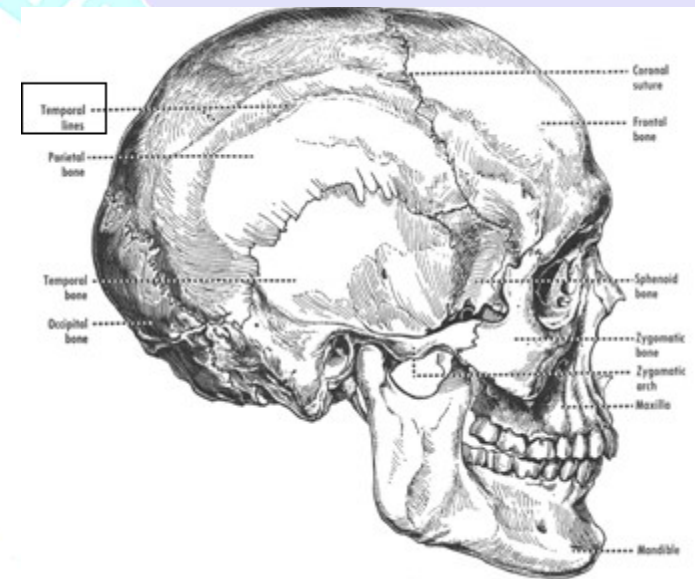
- Pterion – point marked by the earlier position of the anterolateral fontanelle.
- junction of the great wing of the sphenoid bone with the frontal, parietal and temporal bones.
- 3 cm behind and slightly above the zygomatic process of the frontal bone.





TEMPORAL FOSSA.

- Temporal Fossa – “temple”
 - Boundries :
 - posteriorly and superiorly – temporal lines
 - superior temporal line
 - inferior temporal line
 - anteriorly – frontal and zygomatic bones
 - laterally – zygomatic arch
 - inferiorly – infratemporal crest of the sphenoid bone
 - floor – frontal, parietal, temporal and greater wing of the sphenoid
 - roof – temporalis muscle and fascia

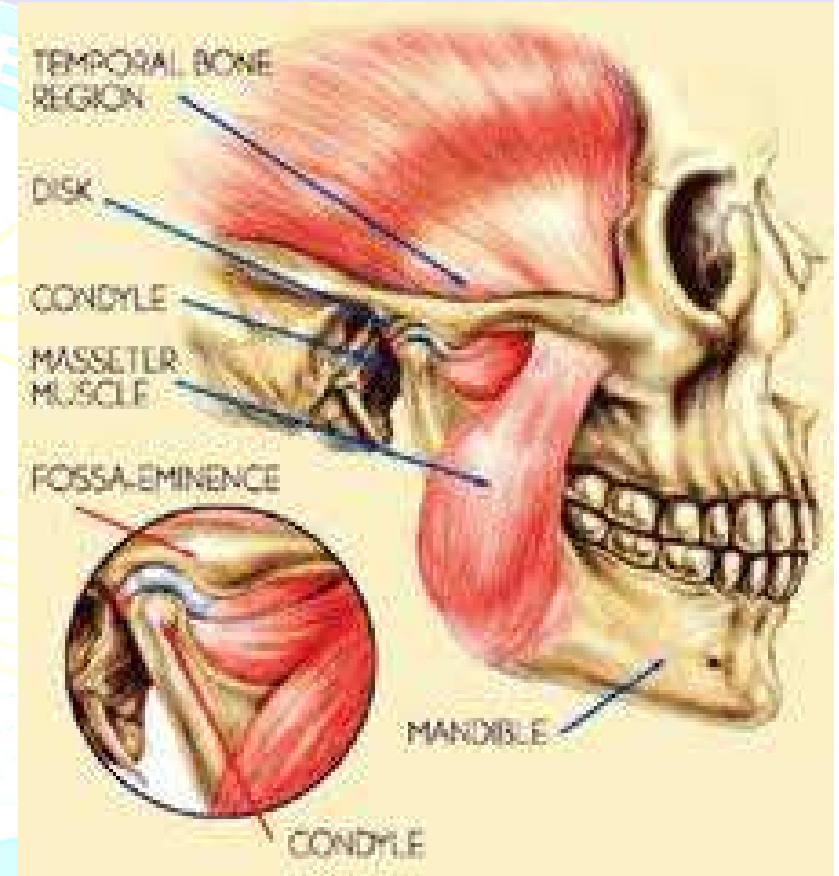


Temporal Fossa.

- Area above the zygomatic arch.
 - Occupied by:
 1. Temporalis and its fascia
 2. Deep temporal nerve and vessels
 3. Auriculo-temporal nerve
 4. Sup. Temporal artery
1. Temporal Fascia
 - Attached:
 - superior to superior temporal line
 - Inserted:
 - Upper border of zygomatic arch
 - Covers the temporalis
 2. Deep temporal nerve
 - From the anterior division of V3
 - Emerge from:
 - upper border of lateral pterygoid to enter temporalis

Contents

- temporal fascia
 - stretches over the temporal fossa and temporalis muscle
 - inferiorly splits into two layers
 - » superficial layer – attaches to the superior margin of the zygomatic arch
 - » deep layer – passes medial to the zygomatic arch to become continuous with the fascia to the masseter muscle
- temporalis muscle.



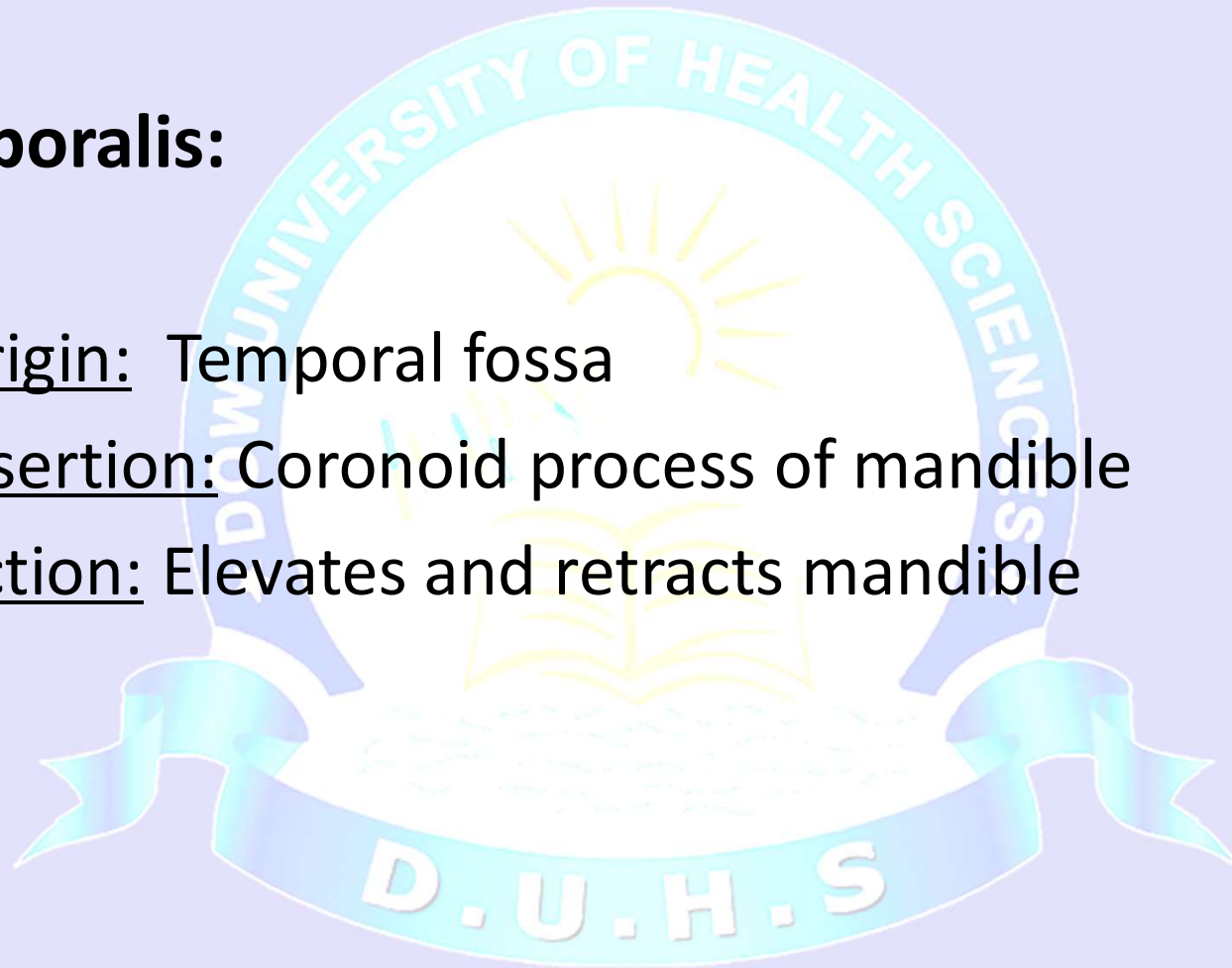
TEMPORAL FOSSA (contd).

- Sutures
 - coronal (frontoparietal).
 - squamosal (parietal-temporal).
 - Frontozygomatic.
 - temporozygomatic .

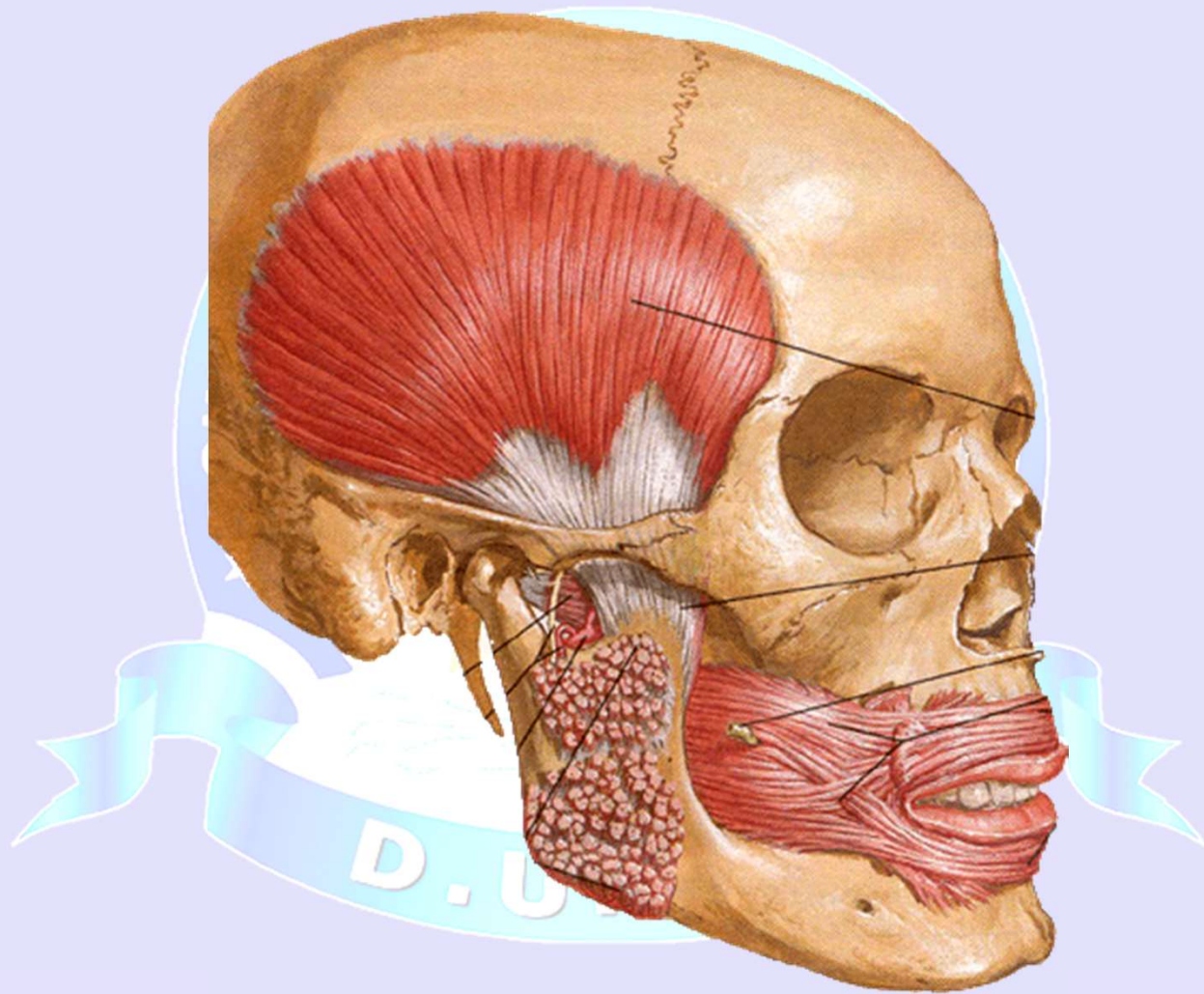


- **Temporalis:**

- Origin: Temporal fossa
- Insertion: Coronoid process of mandible
- Action: Elevates and retracts mandible

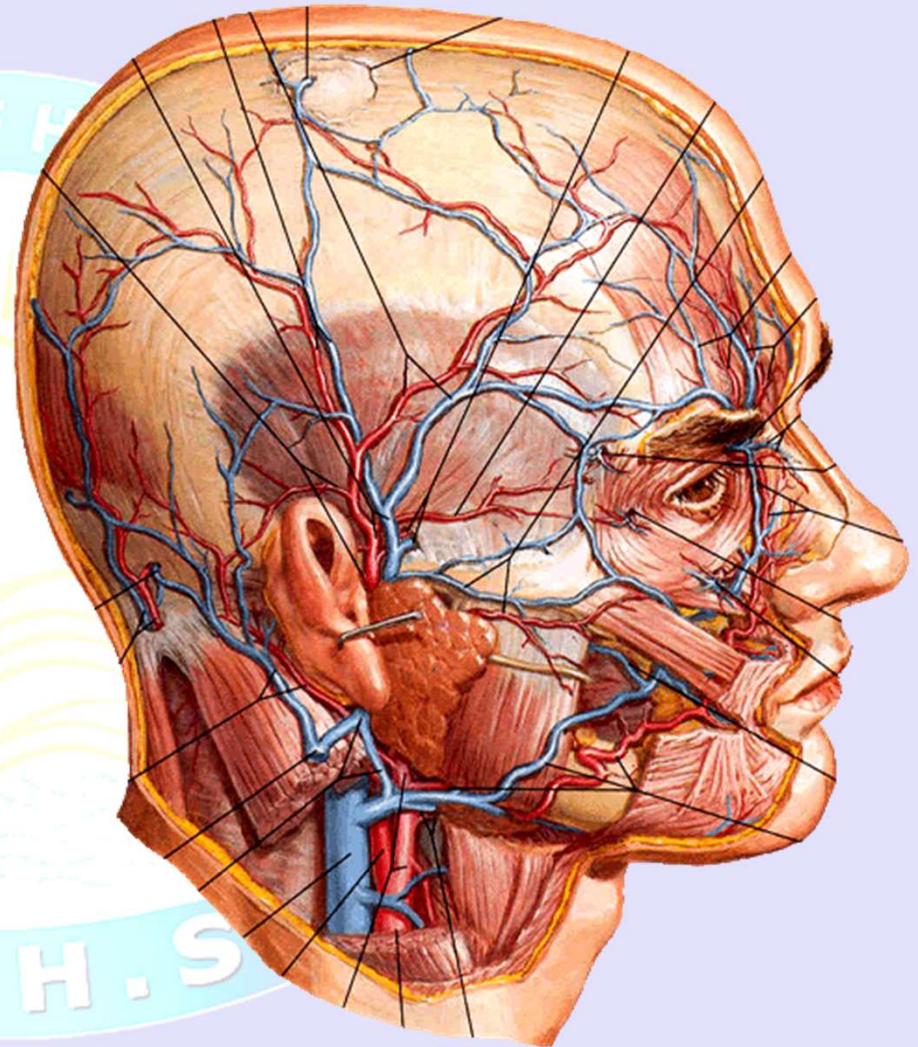


Temporalis



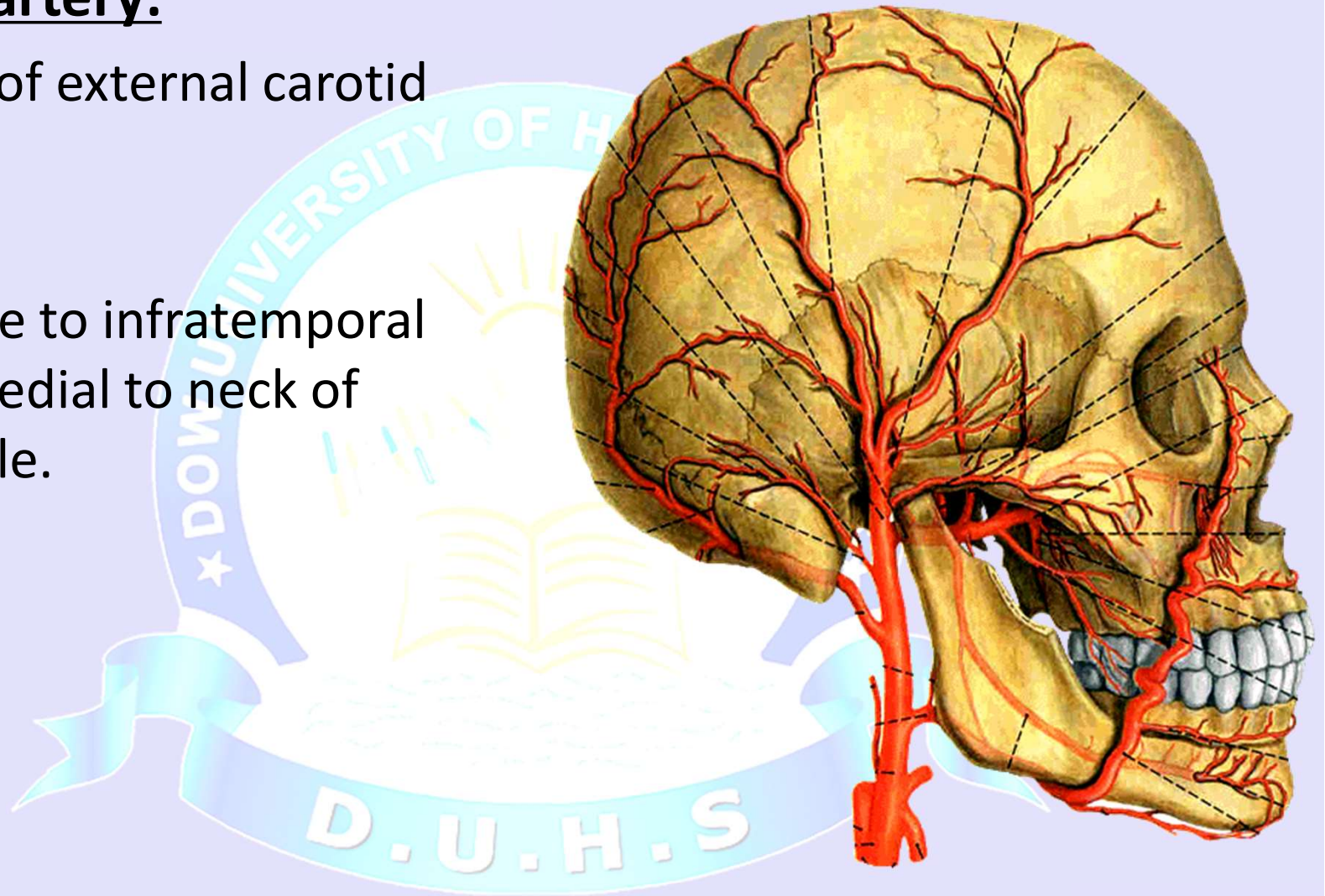
Superficial temporal artery:

- Terminal branch of external carotid a.
- Ascends in front of ear (where it is palpable) to supply temporal and anterior portion of scalp.
- Transverse facial a. runs above parotid duct to supply the cheek region.



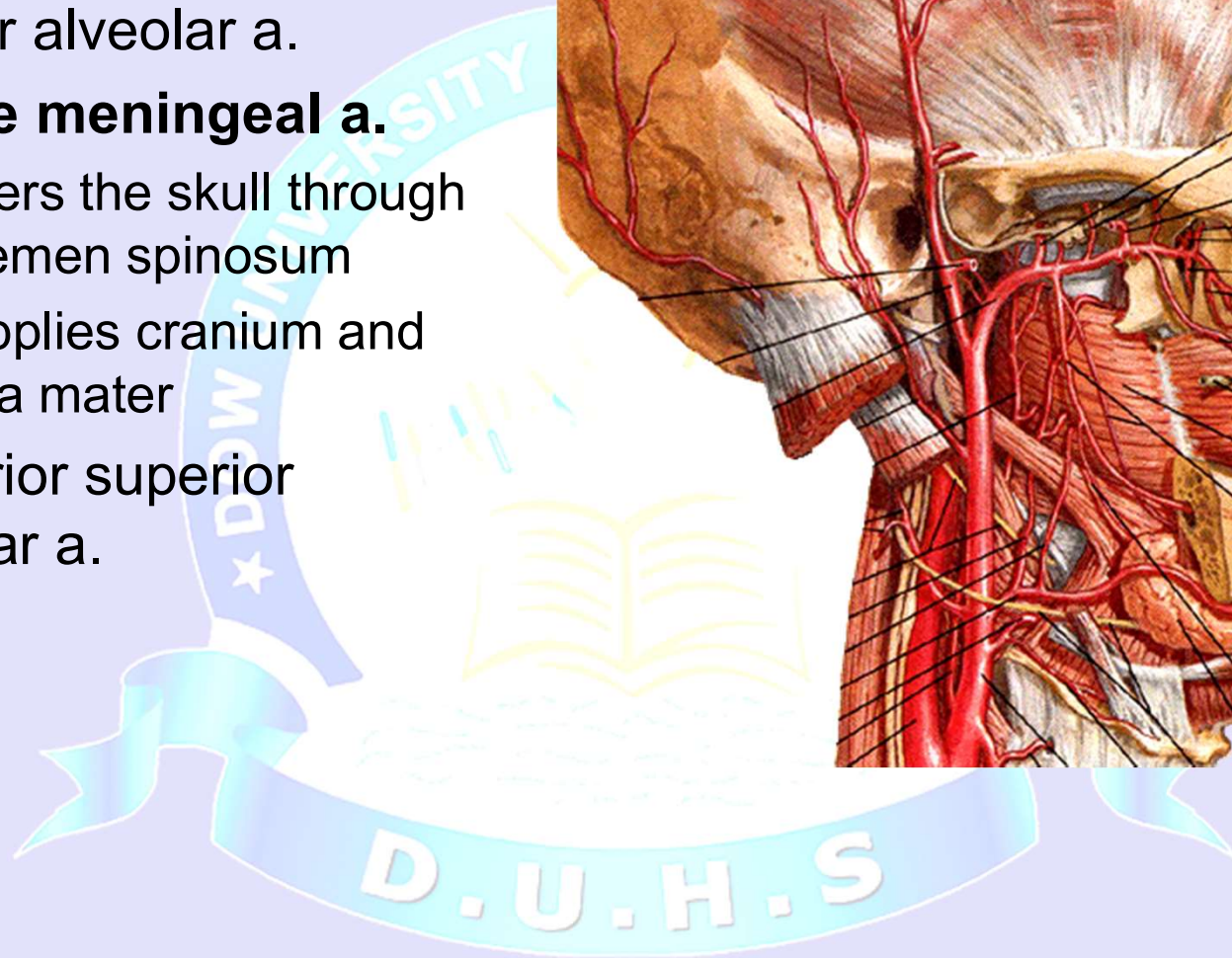
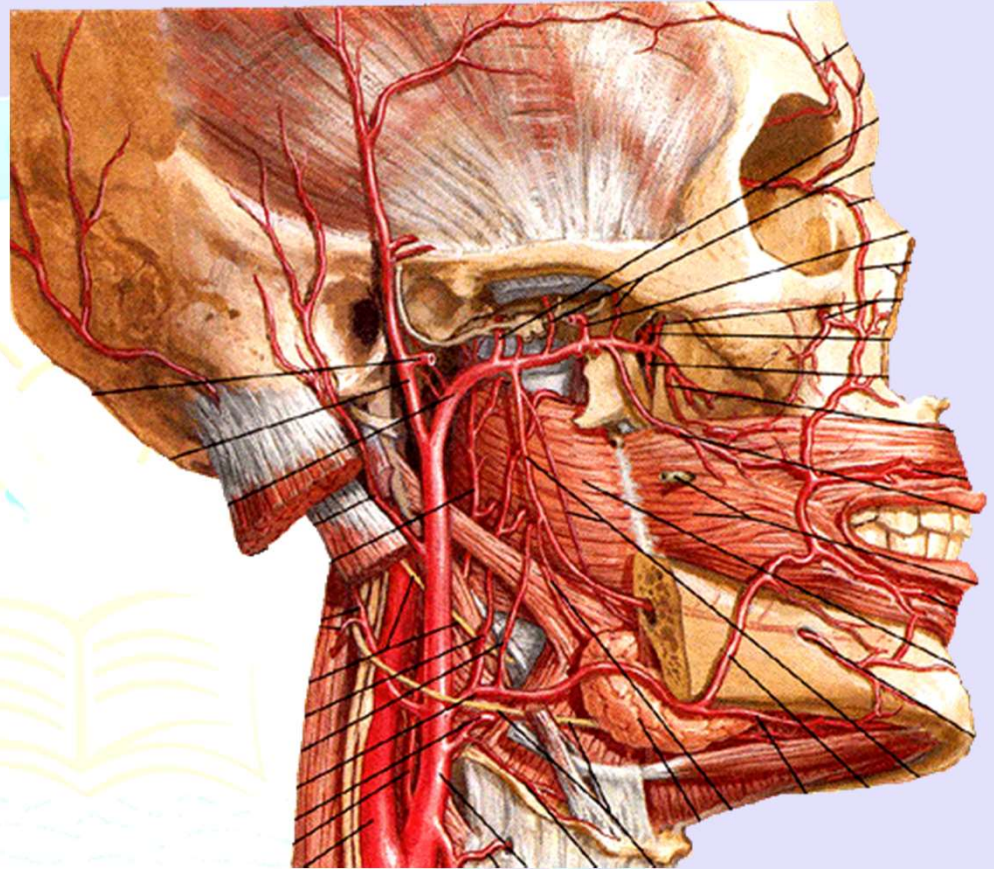
Maxillary artery:

- Branch of external carotid artery.
- Entrance to infratemporal fossa medial to neck of mandible.



- Branches

- Inferior alveolar a.
- **Middle meningeal a.**
 - enters the skull through foramen spinosum
 - Supplies cranium and dura mater
- Posterior superior alveolar a.



Temporomandibular Joint (TMJ).

- The TMJ is a modified hinge type of synovial joint.
- Articular surfaces
 - condyle of the mandible (inferior)
 - articular tubercle of the temporal bone
 - mandibular fossa of the temporal bone (squamous part)
- Articular capsule: The loose, fibrous capsule attaches to the margins of the articular area on the temporal bone and around the neck of the mandible.
 - superior synovial membrane – lines the fibrous capsule superior to the articular disc
 - inferior synovial membrane – lines the capsule inferior to the disc

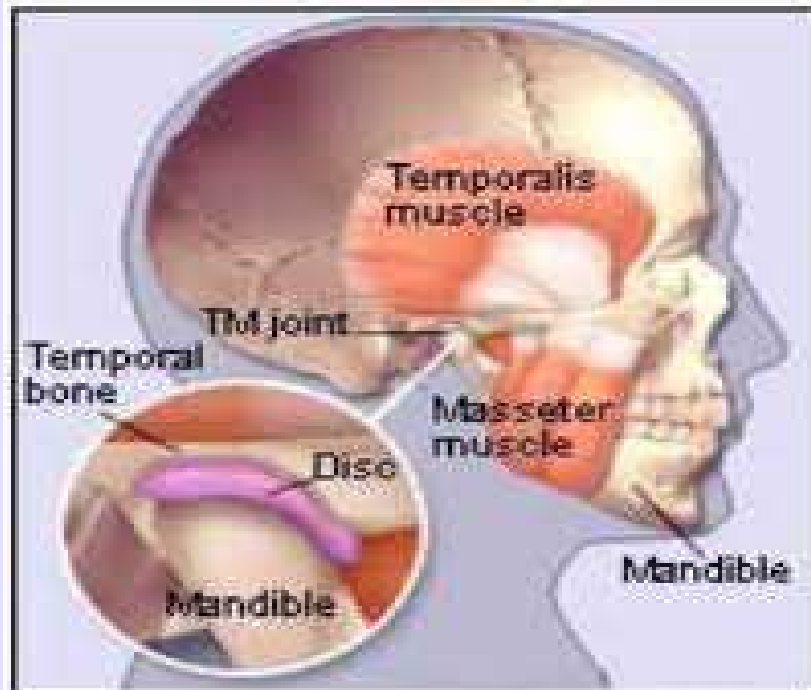
Temporomandibular Joint (TMJ).

- The joint is divided into two compartments by an articular disc – oval, fibrocartilage
 - superior compartment – gliding movements of protrusion and retrusion.
 - inferior compartment – hinge movements of depression and elevation.
 - It is attached more firmly to the mandible than the temporal bone.
 - When the mouth is opened, the disc slides anteriorly on the articular surface against the posterior surface of the articular tubercle.

STABILITY OF THE JOINT

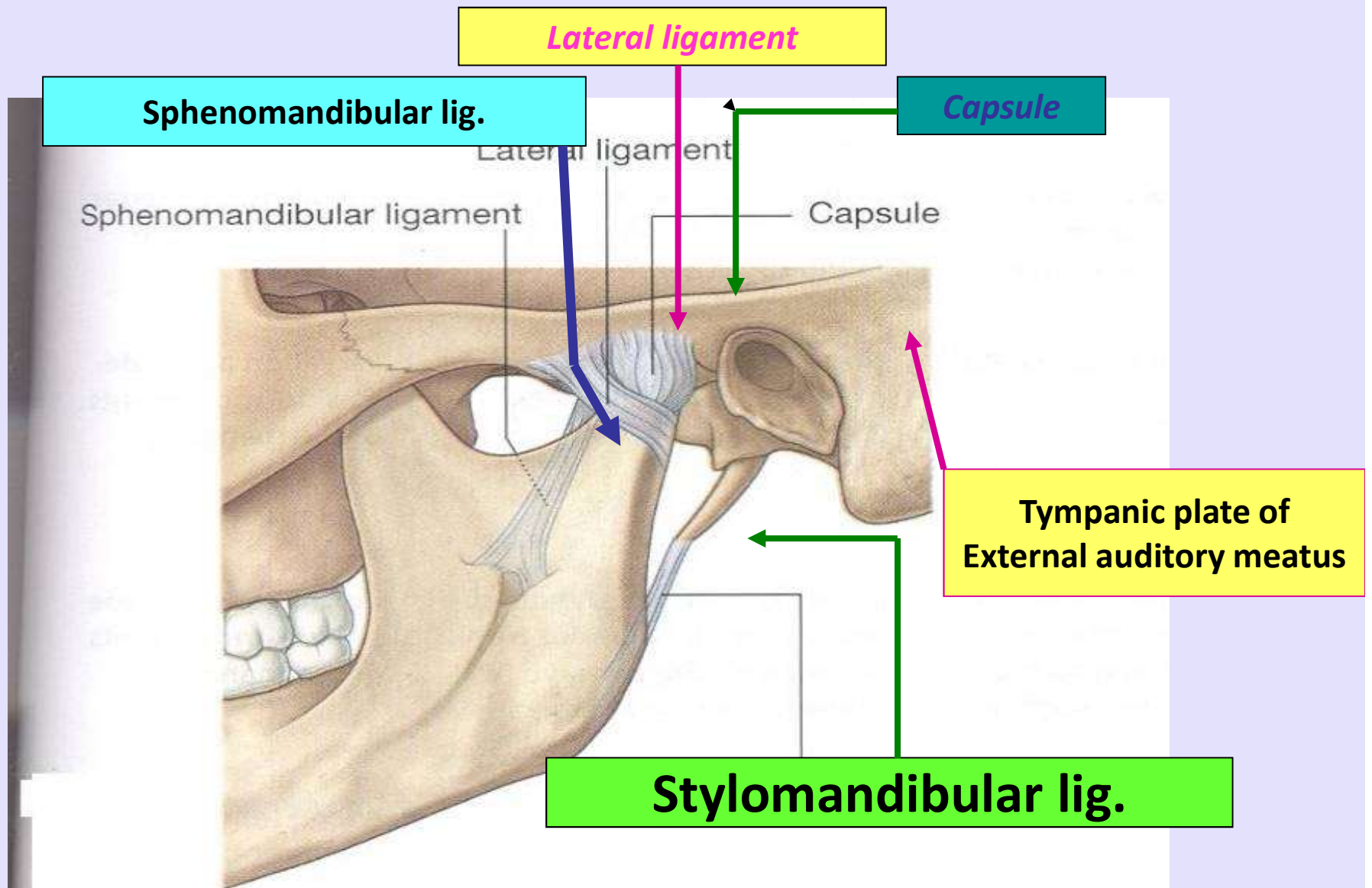
- THE JOINT IS MUCH MORE STABLE WITH THE TEETH IN OCCLUSION THAN WHEN THE JAW IS OPEN

Temporomandibular Joint (TMJ)



LIGAMENTS

- **Intrinsic ligament** - The thick part of the articular capsule forms the intrinsic lateral ligament (TMJ ligament) which strengthens the TMJ laterally.
- Prevents posterior dislocation of the TMJ.
- **Extrinsic ligaments**
 - stylomandibular ligament
 - thickening of the fibrous capsule of the parotid gland
 - extends from the styloid process of the temporal bone to the angle of the mandible
 - does not contribute significantly to the strength of the joint
 - sphenomandibular ligament
 - runs from the spine of the sphenoid to the lingua of the mandible
 - primary passive support of the mandible, although the tonus of the muscles of mastication usually bears the mandible's weight
 - serves as a “swinging hinge” for the mandible for the movements of the mandible at the TMJ



Ligaments of temporomandibular joint

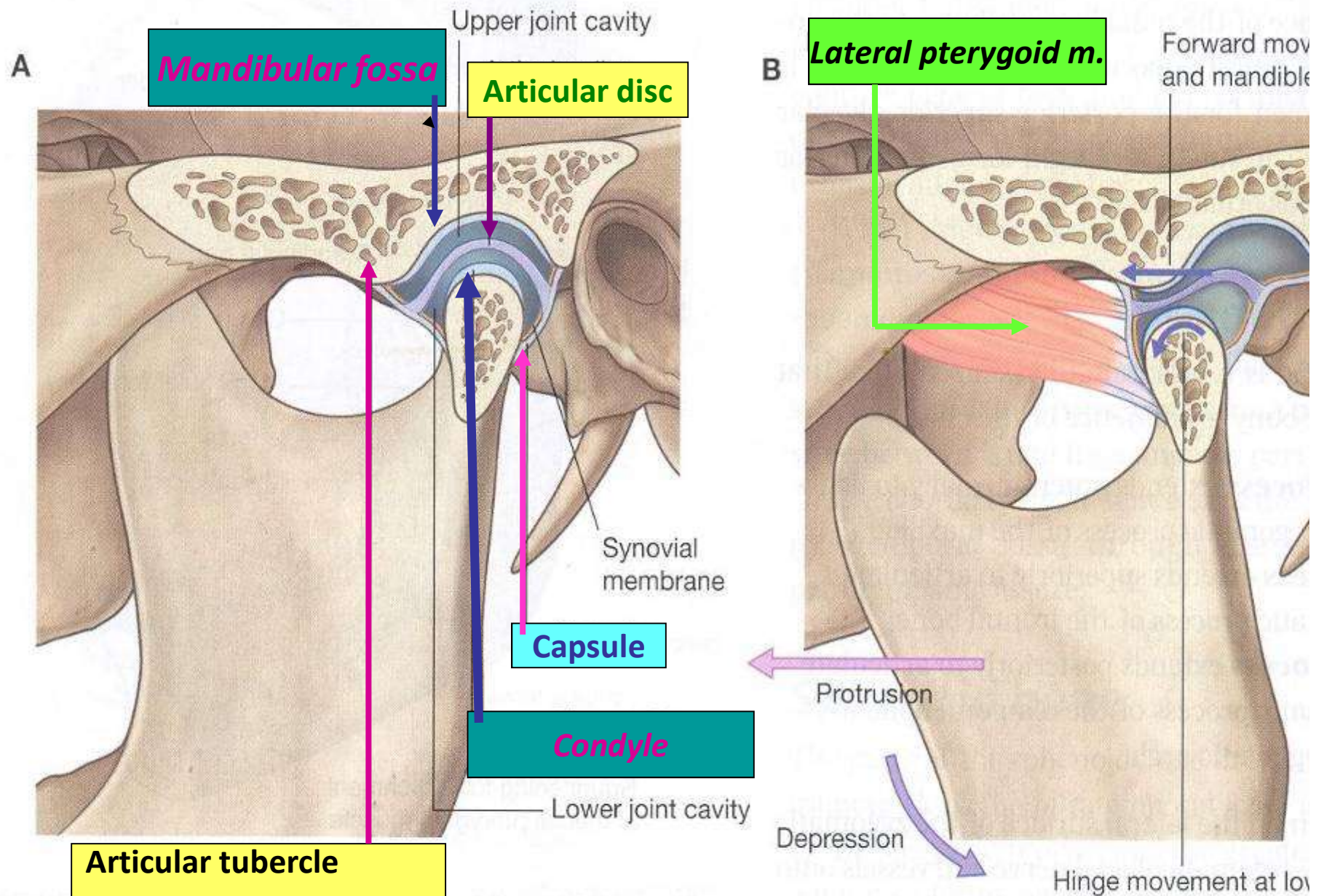


Fig. 29

Temporomandibular joint A.Mouth closed B.Mouth open

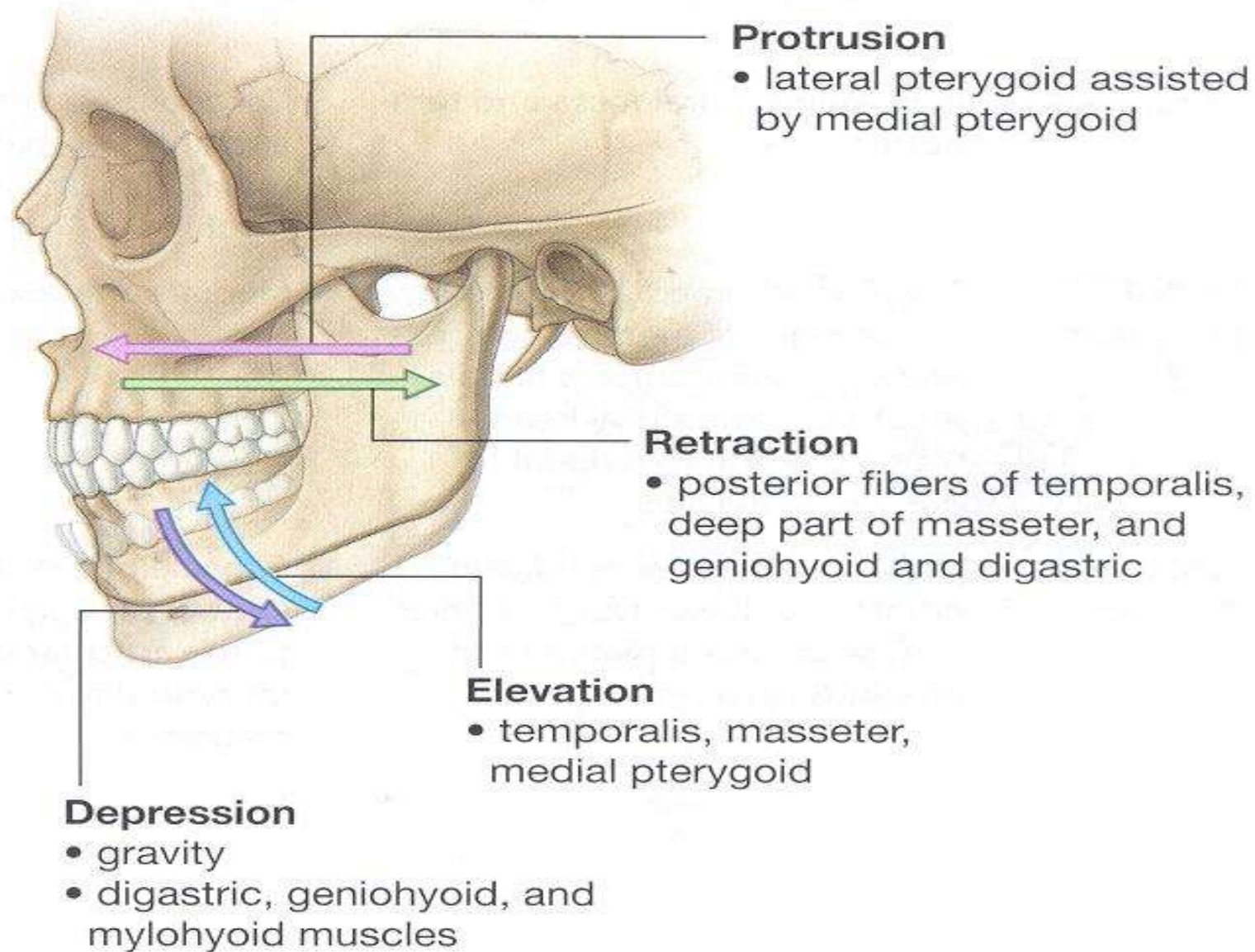
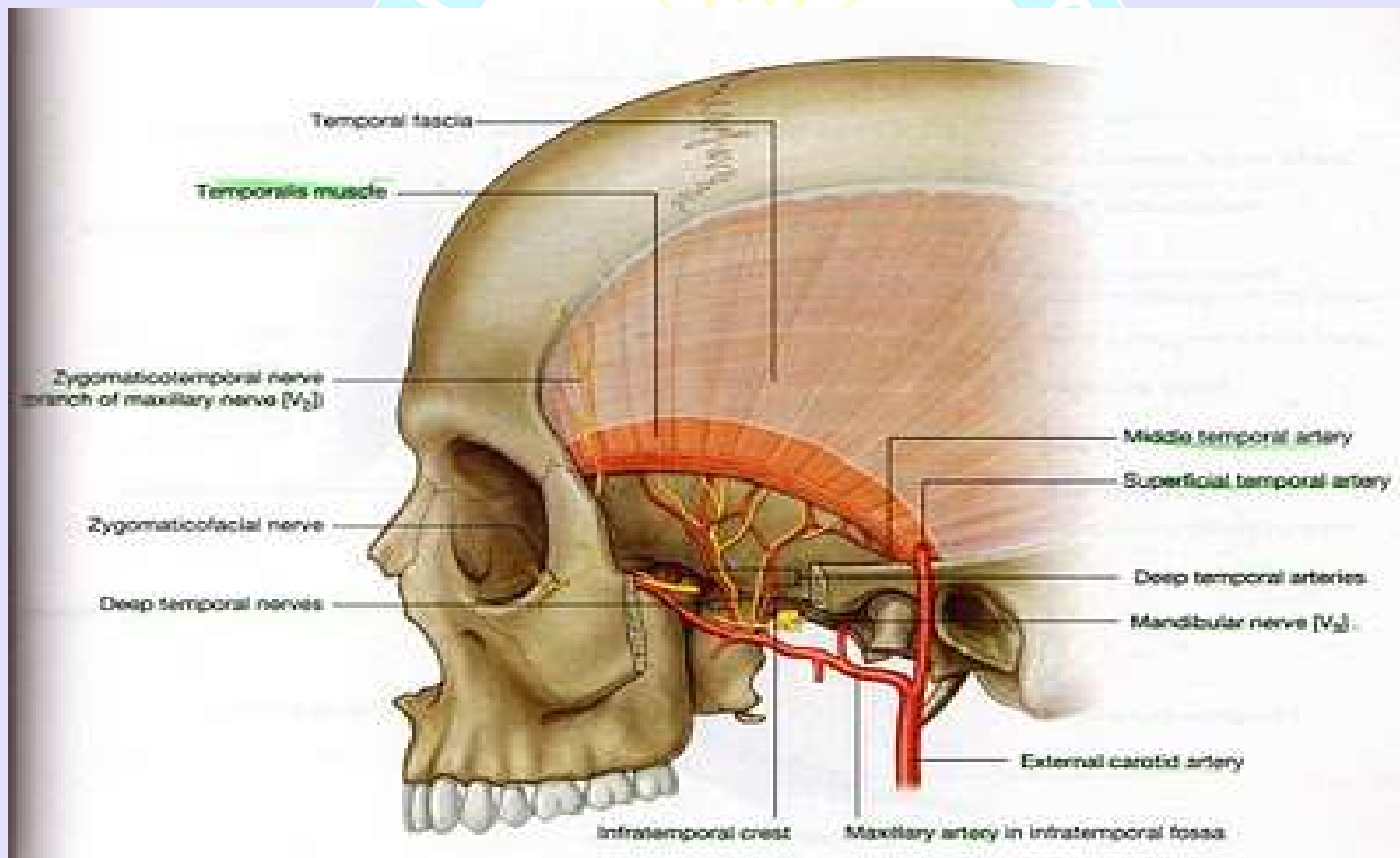


Fig. 31

Movement of temporomandibular joint

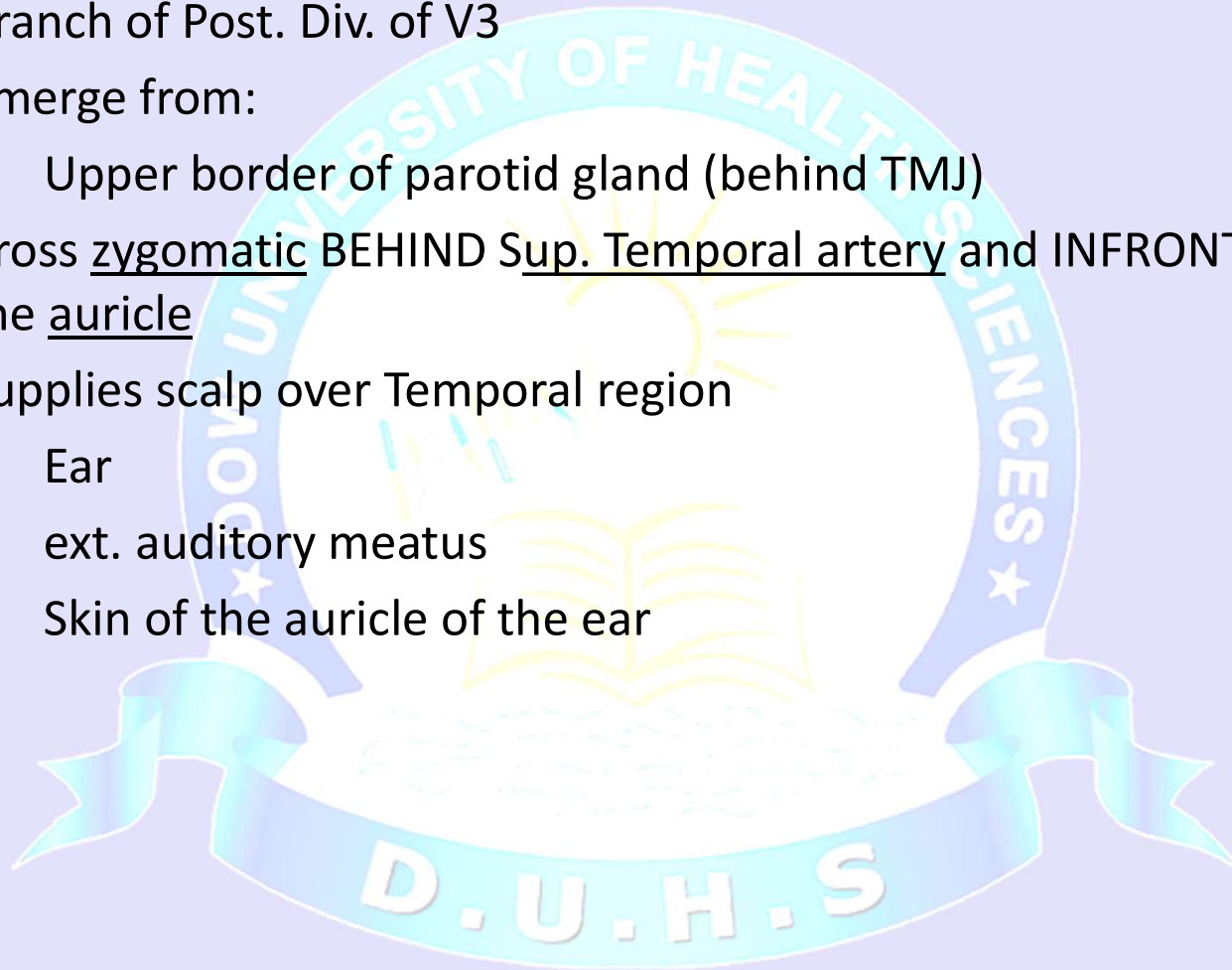
JOINT INNERVATION

- THE NERVE SUPPLY IS DERIVED FROM THE AURICULOTEMPORAL NERVE, ASSISTED BY THE NERVE TO THE MASSETER (HILTON'S LAW)



Auriculotemporal nerve:

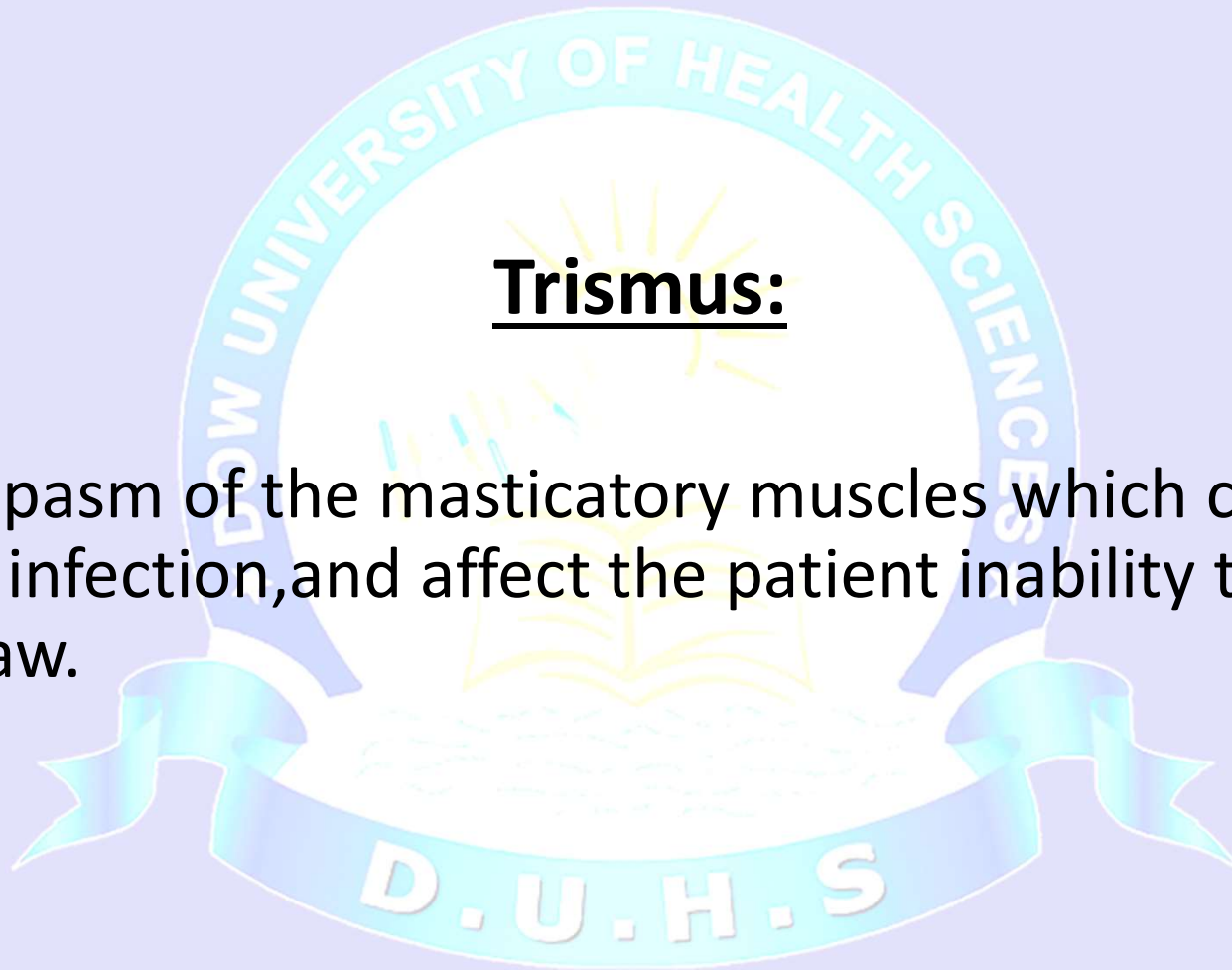
- Branch of Post. Div. of V3
- Emerge from:
 - Upper border of parotid gland (behind TMJ)
- Cross zygomatic BEHIND Sup. Temporal artery and INFRONT of the auricle
- Supplies scalp over Temporal region
 - Ear
 - ext. auditory meatus
 - Skin of the auricle of the ear



CLINICAL CORRELATION

Trismus:

- The spasm of the masticatory muscles which caused from infection, and affect the patient inability to open the jaw.



CLINICAL CORRELATION(con'd)

Tic Dourulaux:

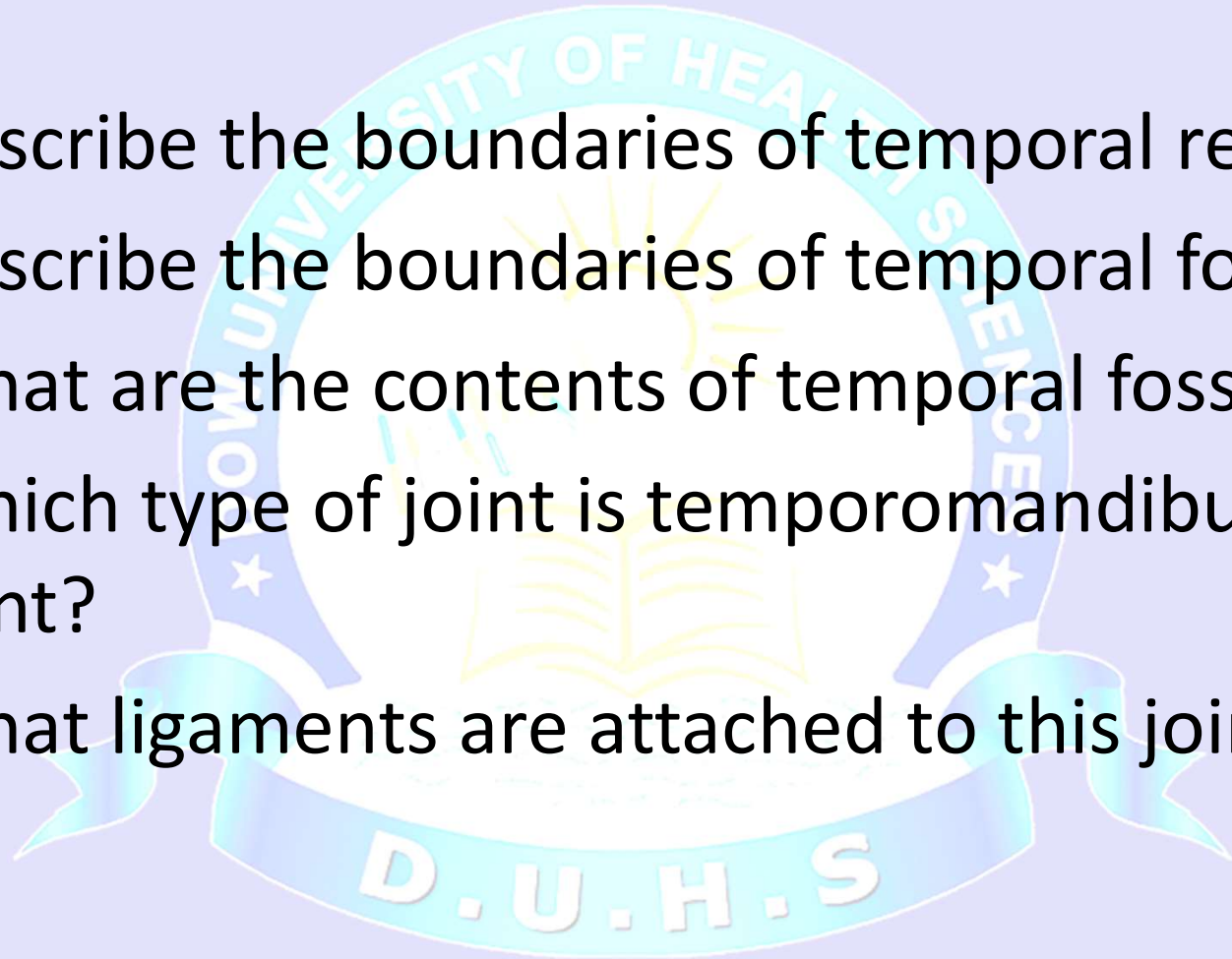
- The excruciating pain along the Trigeminal nerve branches.

Posterior dislocation of the TMJ:

- it is caused from strong force to the anterior part of mandible and affected tympanic plate of the external auditory meatus injury, eventually bleeding could occurs.

SELF ASSESSMENT.

1. Describe the boundaries of temporal region?
2. Describe the boundaries of temporal fossa?
3. What are the contents of temporal fossa?
4. Which type of joint is temporomandibular joint?
5. What ligaments are attached to this joint?





Thank You!